

PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

RescueRoute – Disaster Response and Evacuation Management System

Student / Team Information

Team Name: Team #:	Crises Coders DSCPP-III-2026-T138
Team member 1 (Team Lead)	Kalra,Khushi- 2510380040 2510380040@geu.ac.in 
Team member 2	Agarwal,Ketan- 2510370403 2510370403@geu.ac.in 
Team member 3	Jindal,Kavya- 2510012913 2510012913@geu.ac.in 
Team member 4	Shukla,Bhoomi- 2510010403 2510010403@geu.ac.in 

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

Natural disasters can be very difficult to deal with, and the situation can become life threatening when roads get blocked or damaged. Recent disasters such as earthquakes and other natural calamities in Nepal have shown the destruction caused and the difficulties faced in providing timely aid to affected people. We have also considered patients who need emergency treatment and must be taken to a hospital immediately. Blocked roads, limited ambulances and hospital capacity can make emergency response more difficult. Therefore, this project aims to develop a system that can help manage disasters and evacuation using Data Structures and Object Oriented Programming Concepts.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

The system needs a way to manage routes and emergency requests. In our project, the road network will be treated like a graph using an adjacency list, where locations are treated as nodes and roads as connections between them. Blocked roads or graph paths will be excluded from available routes. We will implement BFS for optimal path finding and use Priority Queues for evacuation and emergency requests. Object Oriented Programming will be used to represent patients, ambulances and hospitals.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

Goal:

Build a C/C++ disaster and evacuation management system to manage locations, roads, emergency requests, patients, ambulances and hospitals. Graphs and BFS will be used for route finding, while Priority Queues will handle emergency cases based on urgency.

Milestones:

Milestone	Work
1. Core Foundation	Create system structure and objects for patients, ambulances, hospitals, disasters and evacuees.
2. Road Network	Build the road network using graphs and manage blocked roads.
3. Route Finding	Use BFS to find available routes between locations.
4. Emergency Priority	Use Priority Queue to handle critical cases first.
5. Record Management	Use Linked Lists, Stacks, Searching and Sorting for records.
6. Disaster Simulation	Simulate roads, patients, emergencies and hospital availability.
7. Testing	Test all modules under different disaster situations.

Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

Because this is a combined Project-Based Learning (PBL) assignment, we will build this project using C and C++ by combining Data Structures with Object Oriented Programming (OOP). The project will work as a disaster and evacuation management system.

OOP Topics Used:

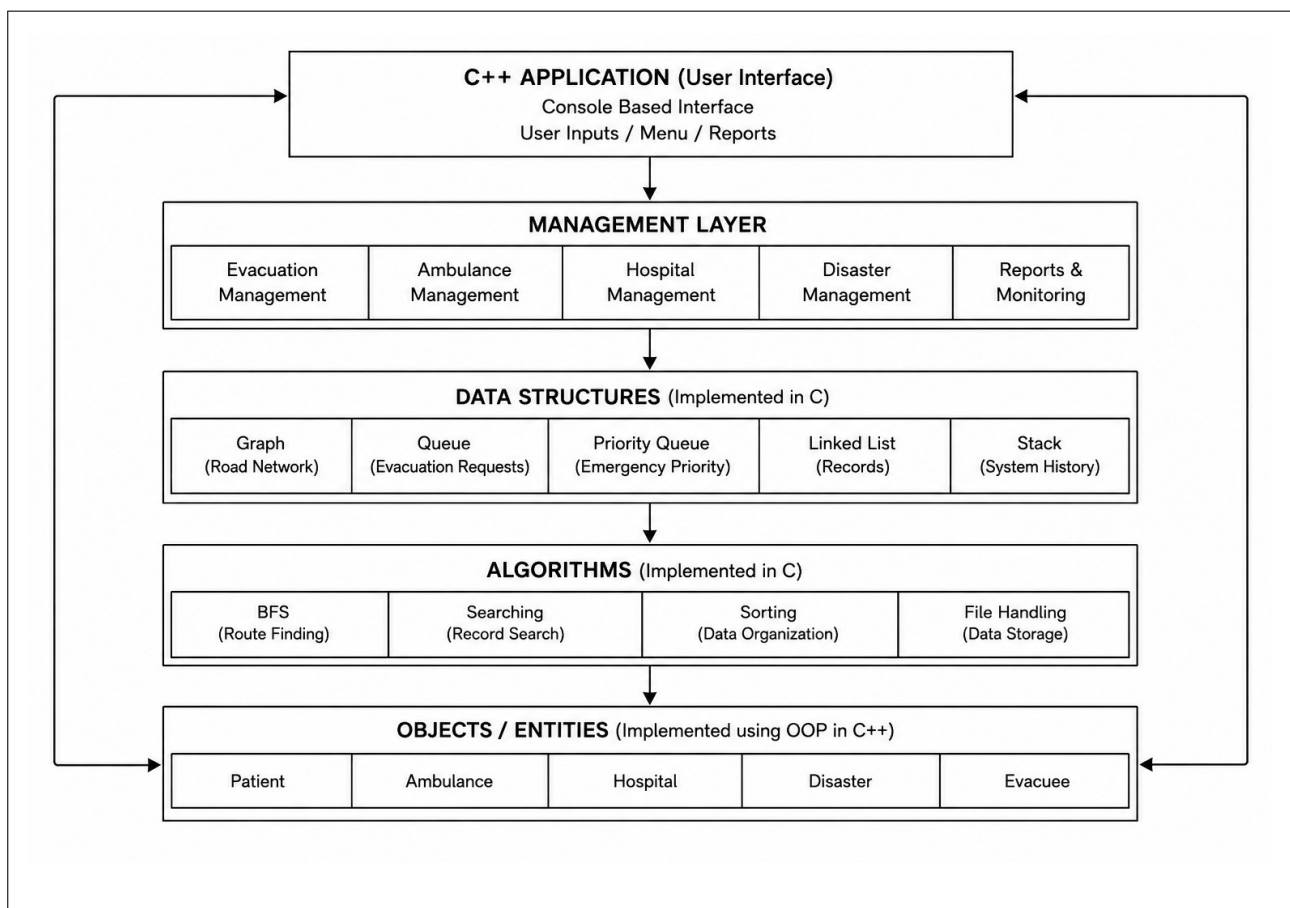
- 1. Classes and Objects:** We will use classes to represent objects such as patients, ambulances, hospitals, disasters and evacuees. Each object will store and manage its related information.
- 2. Encapsulation:** Important information related to patients, hospitals and ambulances will be kept inside their respective classes and accessed through controlled functions.

DS Topics Used:

- 1. Graphs:** Locations will be represented as nodes and roads as connections between them.
- 2. BFS (Breadth-First Search):** BFS will be used to find available routes while blocked roads are excluded.
- 3. Queues:** Queues will manage evacuation and emergency requests in the order received.
- 4. Priority Queues:** Critical emergency cases will receive higher priority and be handled first.
- 5. Linked Lists:** Linked Lists will manage records dynamically as new records are added.
- 6. Stacks:** Stacks will maintain a history of recent system actions.
- 7. Searching and Sorting:** Searching will find specific records and sorting will organize information.
- 8. File Handling:** Files will store important records and information for later access.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

This project will result in a working Disaster and Evacuation Management System using Data Structures in C and Object-Oriented Programming in C++.

1. A working system to register disasters, patients, evacuees, hospitals and ambulances.
2. Graphs and BFS for representing locations and finding available routes when roads are blocked.
3. Queues for evacuation requests and Priority Queues for critical emergency cases.
4. Linked Lists, Stacks, Searching and Sorting techniques for record management, system history and organizing data.
5. C++ classes for Patient, Ambulance, Hospital, Disaster and Evacuee.
6. Disaster simulations demonstrating route finding, ambulance dispatch and hospital allocation.
7. File Handling for saving records, along with complete source code and final project demonstration.

Assumptions

(Describe the assumptions used to solve the problem. Max 100 words).

We assume that the system will use a simplified, unweighted road graph where each road can have a LOW, MEDIUM, HIGH or BLOCKED status. BFS will be used for optimal paths between locations.

Hospital capacity, ambulance availability and patient severity will be stored as basic records and used during the simulation.

The system is designed as a small-scale academic simulation, so real-world factors such as live GPS data, GIS mapping, real-time traffic conditions and cloud services are not considered. Dijkstra's algorithm is also outside the scope, as BFS is sufficient for the simplified road network used in the project.

References

1. Kazi Ashik Islam et al., "A Simulation-based Approach for Large-scale Evacuation Planning," IEEE, 2020.
<https://ieeexplore.ieee.org/Xplore/home.jsp>IEEE Xplore
2. Cormen et al., *Introduction to Algorithms*, MIT Press.
<https://mitpress.mit.edu/9780262046305/introduction-to-algorithms>MIT Press
3. Bjarne Stroustrup, *The C++ Programming Language*.
<https://www.stroustrup.com/C%2B%2B.html>C++ Programming Language